ADENOSINE

Rapid push + flush

Warn: brief impending doom

CHEMICAL NAME (2R,3R,4S,5R)-2-(6-amino-9H-purin-9-yl)-5-(hydroxymethyl)oxolane-3,4-diol

CLASS Endogenous purine nucleoside; class V antiarrhythmic

TRADE NAMES Adenocard · Adenocor · Adenoscan

THERAPEUTIC USES / INDICATIONS

- Termination of paroxysmal SVT (AVNRT, AVRT) after vagal manoeuvres fail.
- Diagnosis of narrow/broad complex tachycardia (unmasks atrial flutter).
- Pharmacological stress testing (myocardial perfusion imaging).

ADMINISTRATION

- **IV rapid bolus over 1–2 seconds into a large proximal vein (antecubital)**, immediately followed by 20 mL saline flush.
- Patient on continuous ECG monitoring; record rhythm strip.
- Defibrillator and resuscitation equipment at bedside.

DOSAGES

- **Adult:** 6 mg rapid IV; if no response after 1–2 min, 12 mg, then a further 12 mg (some guidelines 18 mg).
- **Central line / heart transplant / dipyridamole:** start 3 mg.
- **Children:** 0.1 mg/kg (max 6 mg), then 0.2 mg/kg (max 12 mg).

PHARMACOKINETICS

- **Onset:** seconds. **Duration:** <10 seconds.
- Rapidly taken up by red cells and endothelium; deaminated.
- **t½:** <10 seconds.

PHARMACODYNAMICS

- Stimulates **A1 receptors** in AV node → ↑ K^+ efflux → transient AV block → interrupts re-entry circuits.
- A2 receptors → coronary vasodilation (stress testing).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Chest discomfort/tightness, sense of impending doom, flushing.
- Dyspnoea, headache, nausea.
- Transient asystole/bradycardia for a few seconds (expected).

ADVERSE EFFECTS (SERIOUS)

- Bronchospasm (asthma).
- Prolonged asystole or AV block (heart transplant, dipyridamole).
- AF induction; ventricular arrhythmias.
- Accelerated conduction in WPW with AF → VF.

CONTRAINDICATIONS & CAUTIONS

- Asthma or severe COPD with bronchospasm.
- 2nd/3rd-degree AV block, sick sinus syndrome (without pacemaker).
- Long QT; severe hypotension; decompensated heart failure.
- Pre-excited AF/flutter (WPW).

KEY INTERACTIONS

- **Dipyridamole:** potentiates — use 3 mg.
- **Theophylline, aminophylline, caffeine:** antagonise; higher dose may be needed.
- **Carbamazepine:** ↑ heart block.

NURSING MANAGEMENT

- Ensure continuous ECG and recording strip running.
- Two-syringe or three-way tap technique: drug then fast flush.
- Warn patient about brief chest tightness and feeling of doom.
- Monitor BP, HR and rhythm after conversion.
- Have defibrillator and atropine ready.

PATIENT EDUCATION

- You may feel a brief, strong chest pressure and flushing for a few seconds — this passes quickly.
- Learn vagal manoeuvres (modified Valsalva) for future episodes.

KEY POINTS

- Transient AV block; terminates SVT.
- t½ <10 s.
- 6 mg → 12 mg → 12 mg rapid push + flush.
- Avoid in asthma; dipyridamole potentiates, theophylline blocks.